STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Define centre of mass. [1]
2. Torque magnitude? [1]
3. SI unit of moment of inertia? [1]
4. Rotational kinetic energy? [1]
5. Condition for angular momentum conservation? [1]
6. Rolling relation? [1]
7. Static rigid-body equilibrium? [1]
8. Parallel-axis theorem? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. 2 and 3 kg at x=0 and 5 m: x_cm? [2]
10. 4 N perpendicular at .5 m: torque? [2]
11. I=3 kg m^2, omega=4: rotational energy? [2]
12. Radius .4 m wheel at 8 m/s: omega? [2]
13. Torque 12 N m and I=4: ? [2]
14. No external force: total momentum? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. 2 kg at x=0 and 6 kg at x=4: x_cm? [3]
16. .25 m spanner, 40 N at 30 deg : torque? [3]
17. Solid disc M=4 kg, R=.5 m: I and KE at omega=10? [3]
18. omega rises 2 to 10 rad/s in 4 s: and theta? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. 2 kg at x=0 and 3 kg at x=5 m, external force 10 N: x_cm and acceleration? [4]
20. Skater I changes 4 to 1.6 kg m^2 from omega=3: final omega and KE change? [4]

End of paper - check working, units, signs and significant figures.